

```

> restart: unprotect('D'):

m := 8;

twist := - 7;

d := 12;                degree := d + 1:

##

HSS := T^13 + T^6*X^7 + X^13:

##

SS := subs({T=1,X=x}, HSS);

##

Jet_xR := expand(

+ (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j *
Delta_xR^0, j=0..m-0*3)

+ (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j *
Delta_xR^1, j=0..m-1*3)

+ (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j *
Delta_xR^2, j=0..m-2*3)

+ (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j *
Delta_xR^3, j=0..m-3*3)

+ (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j *
Delta_xR^4, j=0..m-4*3) ):

##

for l from 0 to m - 0*3 do dA[l] := (m-0*2)*d - m + l*1 + 0*1 +
twist: od:

for l from 0 to m - 1*3 do dB[l] := (m-1*2)*d - m + l*1 + 1*1 +
twist: od:

for l from 0 to m - 2*3 do dC[l] := (m-2*2)*d - m + l*1 + 2*1 +
twist: od:

```

```
for l from 0 to m - 3*3 do dD[l] := (m-3*2)*d - m + l*1 + 3*1 +
twist: od:
```

```
for l from 0 to m - 4*3 do dE[l] := (m-4*2)*d - m + l*1 + 4*1 +
twist: od:
```

```
##
```

```
Delta_xR := x1 * (R2/RR - R1*R1/RR^2) - x2 * (R1/RR):
```

```
Solx1 := - y1*Ry/Rx + R1/Rx:
```

```
Solx2 := - Rxx*y1^2*Ry^2/Rx^3 + 2*Rxx*y1*Ry*R1/Rx^3 - Rxx*
R1^2/Rx^3 + 2*y1^2*Rxy*Ry/Rx^2
- 2*y1*Rxy*R1/Rx^2 - y1^2*Ryy/Rx - y2*Ry/Rx + R2/Rx:
```

```
##
```

```
Jet_Ry := sort(mtaylor(
```

```

            expand(subs(R2=-(RR^2*Delta_Ry-RR*R1*y2-
R1^2*y1)/(RR*y1),
            expand(subs({x1=Solx1,x2=Solx2},
            Jet_xR))))),
```

```

[R1,y1,Delta_Ry], 100), [R1,y1,Delta_Ry]):
```

```
##
```

```
printlevel := 2:
```

```
for k from 0 to m/3 do
```

```
for j from 0 to m-3*k do
```

```
EEq[m-j-3*k,j,k] := factor(
```

```

    Ry^(m-j-3*k)*RR^(m-j-3*k)*Rx^(m-2*k)
    *
```

```

    coeftayl(Jet_Ry, [R1,y1,Delta_Ry]=[0,0,0], [m-j-3*k,j,k]) ):

```

```
od: od:
```

```
## EQUATION RR EXPLICITE
```

```
RR := y^(d+1) + SS;
```

```
HRR := Y^(d+1) + HSS;
```

```
RY := diff(HRR, Y):
```

```

Rx := diff(RR,x):
Ry := diff(RR,y):

Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Ryy := diff(RR,y,y):

##

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], y, (m-j-3*k)*d):

Var[m-j-3*k,j,k] := indets(Eq[m-j-3*k,j,k]) minus {x,y}:

od: od:

##

```

$$\begin{aligned}
 m &:= 8 \\
 \text{twist} &:= -7 \\
 d &:= 12 \\
 SS &:= x^{13} + x^7 + 1 \\
 RR &:= x^{13} + y^{13} + x^7 + 1 \\
 HRR &:= T^{13} + T^6 X^7 + X^{13} + Y^{13}
 \end{aligned}$$

(1)

```

> ## EXPAND ALL UNKNOWNNS UP TO y^(1*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],1*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],1*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],1*d-1)): od:

for l from 0 to m - 3*3 do

```

```

DD||I := add(add(D||I[i,j]*x^i*y^j, i=0..dD[I]-j), j=0..min(dD
[I],1*d-1)): od:

for l from 0 to m - 4*3 do
EE||I := add(add(E||I[i,j]*x^i*y^j, i=0..dE[I]-j), j=0..min(dE
[I],1*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(1*d)

Systemey1d := {}:

for k from 0 to (m-1)/3 do

Coeffs_y1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,
k], y, 1*d)), [x,y])}:

Systemey1d := Systemey1d union Coeffs_y1d[1,m-1-3*k,k]:

od:

Systeme_y1d := Systemey1d:

## VARIABLES FOR DIVISIBILITY BY y^(1*d)

Variables_y1d := indets(Systeme_y1d):

## RESOLUTION y^(1*d)

Solutions_y1d := solve(Systeme_y1d, Variables_y1d):

assign(Solutions_y1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||I)) od;
for l from 0 to m - 1*3 do nops(expand(BB||I)) od;
for l from 0 to m - 2*3 do nops(expand(CC||I)) od;
for l from 0 to m - 3*3 do nops(expand(DD||I)) od;
for l from 0 to m - 4*3 do nops(expand(EE||I)) od;

```

4766

13531

942

954

966

978
990
1002
1014
3566
8263
666
678
690
702
1342
2719
390

(2)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(2*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],2*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],2*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],2*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],2*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],2*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(2*d)}$ 

Systemey2d := {}:

for k from 0 to (m-2)/3 do
```

```
Coeffs_y2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], y, 2*d)), [x,y]))}:
```

```
Systemey2d := Systemey2d union Coeffs_y2d[2,m-2-3*k,k]:
```

```
od:
```

```
Systeme_y2d := Systemey2d:
```

```
## VARIABLES FOR DIVISIBILITY BY  $y^{(2*d)}$ 
```

```
Variables_y2d := indets(Systeme_y2d):
```

```
## RESOLUTION  $y^{(2*d)}$ 
```

```
Solutions_y2d := solve(Systeme_y2d, Variables_y2d):
```

```
assign(Solutions_y2d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
```

```
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
```

```
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
```

```
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

```
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

15011

25780

54446

1764

1788

1812

1836

1860

1884

7527

14412

26583

1212

1236

1260

755

1363

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(3*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],3*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],3*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],3*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],3*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],3*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(3*d)}$ 

Systemey3d := {}:

for k from 0 to (m-3)/3 do

Coeffs_y3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], y, 3*d)), [x,y])}:

Systemey3d := Systemey3d union Coeffs_y3d[3,m-3-3*k,k]:

od:

Systeme_y3d := Systemey3d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(3*d)}$ 

Variables_y3d := indets(Systeme_y3d):

## RESOLUTION  $y^{(3*d)}$ 

Solutions_y3d := solve(Systeme_y3d, Variables_y3d):

```

```
assign(Solutions_y3d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;  
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

66877

51466

69391

80765

2466

2502

2538

2574

2610

20903

19564

21555

15135

1638

1674

919

1426

2457

(4)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(4*d)}$ 
```

```
for l from 0 to m - 0*3 do
```

```
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA  
[l],4*d-1)): od:
```

```
for l from 0 to m - 1*3 do
```

```
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB  
[l],4*d-1)): od:
```

```
for l from 0 to m - 2*3 do
```



```

CC||I := add(add(C||I[i,j]*x^i*y^j, i=0..dC[I]-j), j=0..min(dC
[I],4*d-1)): od:

for l from 0 to m - 3*3 do
DD||I := add(add(D||I[i,j]*x^i*y^j, i=0..dD[I]-j), j=0..min(dD
[I],4*d-1)): od:

for l from 0 to m - 4*3 do
EE||I := add(add(E||I[i,j]*x^i*y^j, i=0..dE[I]-j), j=0..min(dE
[I],4*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(4*d)

Systemey4d := {}:

for k from 0 to (m-4)/3 do

Coeffs_y4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], y, 4*d)), [x,y])}:

Systemey4d := Systemey4d union Coeffs_y4d[4,m-4-3*k,k]:

od:

Systeme_y4d := Systemey4d:

## VARIABLES FOR DIVISIBILITY BY y^(4*d)

Variables_y4d := indets(Systeme_y4d):

## RESOLUTION y^(4*d)

Solutions_y4d := solve(Systeme_y4d, Variables_y4d):

assign(Solutions_y4d):

##

for l from 0 to m - 0*3 do nops(expand(AA||I)) od;
for l from 0 to m - 1*3 do nops(expand(BB||I)) od;
for l from 0 to m - 2*3 do nops(expand(CC||I)) od;
for l from 0 to m - 3*3 do nops(expand(DD||I)) od;
for l from 0 to m - 4*3 do nops(expand(EE||I)) od;

```

68885

66789

70519

80072

107683
 3048
 3096
 3144
 3192
 13305
 19298
 22010
 25124
 29121
 1944
 3235
 5018
 7806

(5)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(5*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],5*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],5*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],5*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],5*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],5*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(5*d)}$ 

Systemey5d := {}:

for k from 0 to (m-5)/3 do
```

```
Coeffs_y5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,  
k], y, 5*d)), [x,y])}:  

```

```
Systemey5d := Systemey5d union Coeffs_y5d[5,m-5-3*k,k]:  

```

```
od:  

```

```
Systeme_y5d := Systemey5d:  

```

```
## VARIABLES FOR DIVISIBILITY BY  $y^{(5*d)}$   

```

```
Variables_y5d := indets(Systeme_y5d):  

```

```
## RESOLUTION  $y^{(5*d)}$   

```

```
Solutions_y5d := solve(Systeme_y5d, Variables_y5d):  

```

```
assign(Solutions_y5d):  

```

```
##  

```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;  
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;  

```

45071

51876

55255

56479

71371

70620

3510

3570

3630

436

1181

1782

1928

1864

275

847
1247
1653

(6)

```
> ## EXPAND ALL UNKNOWNNS UP TO y^(6*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],6*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],6*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],6*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],6*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],6*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(6*d)

Systemey6d := {}:

for k from 0 to (m-6)/3 do

Coeffs_y6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,
k], y, 6*d)), [x,y])}:

Systemey6d := Systemey6d union Coeffs_y6d[6,m-6-3*k,k]:

od:

Systeme_y6d := Systemey6d:

## VARIABLES FOR DIVISIBILITY BY y^(6*d)

Variables_y6d := indets(Systeme_y6d):
```

```
## RESOLUTION  $y^{(6*d)}$ 
```

```
Solutions_y6d := solve(Systeme_y6d, Variables_y6d):
```

```
assign(Solutions_y6d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;  
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

9907

15123

19315

21944

24953

26297

25121

3852

3924

690

1658

2500

2902

3010

547

1310

1919

2634

(7)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(7*d)}$ 
```

```
for l from 0 to m - 0*3 do
```

```
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA  
[l],7*d-1)): od:
```

```
for l from 0 to m - 1*3 do
```

```
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
```

```

[l],7*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],7*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],7*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],7*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(7*d)

Systemey7d := {}:

for k from 0 to (m-7)/3 do

Coeffs_y7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
k], y, 7*d)), [x,y])}:

Systemey7d := Systemey7d union Coeffs_y7d[7,m-7-3*k,k]:

od:

Systeme_y7d := Systemey7d:

## VARIABLES FOR DIVISIBILITY BY y^(7*d)

Variables_y7d := indets(Systeme_y7d):

## RESOLUTION y^(7*d)

Solutions_y7d := solve(Systeme_y7d, Variables_y7d):

assign(Solutions_y7d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

160
351
637
820
943
918
360
4074
72
288
467
654
645
103
244
324
565

(8)

>

EXPAND ALL UNKNOWNNS UP TO $y^{(8*d)}$

for l from 0 to m - 0*3 do

AA||l := add(add(A||l[i,j]*xⁱ*y^j, i=0..dA[l]-j), j=0..min(dA[l],8*d-1)): od:

for l from 0 to m - 1*3 do

BB||l := add(add(B||l[i,j]*xⁱ*y^j, i=0..dB[l]-j), j=0..min(dB[l],8*d-1)): od:

for l from 0 to m - 2*3 do

CC||l := add(add(C||l[i,j]*xⁱ*y^j, i=0..dC[l]-j), j=0..min(dC[l],8*d-1)): od:

for l from 0 to m - 3*3 do

DD||l := add(add(D||l[i,j]*xⁱ*y^j, i=0..dD[l]-j), j=0..min(dD[l],8*d-1)): od:

for l from 0 to m - 4*3 do

EE||l := add(add(E||l[i,j]*xⁱ*y^j, i=0..dE[l]-j), j=0..min(dE[l],8*d-1)): od:

```

## SYSTEM FOR DIVISIBILITY BY  $y^{(8*d)}$ 

Systemey8d := {}:

for k from 0 to (m-8)/3 do

Coeffs_y8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,
k], y, 8*d)), [x,y])}:

Systemey8d := Systemey8d union Coeffs_y8d[8,m-8-3*k,k]:

od:

Systeme_y8d := Systemey8d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(8*d)}$ 

Variables_y8d := indets(Systeme_y8d):

## RESOLUTION  $y^{(8*d)}$ 

Solutions_y8d := solve(Systeme_y8d, Variables_y8d):

assign(Solutions_y8d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

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(9)

```
> ## EXPAND ALL UNKNOWNNS UP TO y^(9*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],9*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],9*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],9*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],9*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],9*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(9*d)

Systemey9d := {}:

for k from 0 to (m-9)/3 do

Coeffs_y9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,
k], y, 9*d)), [x,y])}:

Systemey9d := Systemey9d union Coeffs_y9d[9,m-9-3*k,k]:

od:

Systeme_y9d := Systemey9d:
```

```

## VARIABLES FOR DIVISIBILITY BY  $y^{(9*d)}$ 

Variables_y9d := indets(Systeme_y9d):

## RESOLUTION  $y^{(9*d)}$ 

Solutions_y9d := solve(Systeme_y9d, Variables_y9d):

assign(Solutions_y9d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EF||l)) od;

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(10)

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>
## EXPAND ALL UNKNOWNNS UP TO  $y^{(10*d)}$ 

for l from 0 to m - 0*3 do

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AA||I := add(add(A||I[i,j]*x^i*y^j, i=0..dA[I]-j), j=0..min(dA
[I],10*d-1)): od:

for l from 0 to m - 1*3 do
BB||I := add(add(B||I[i,j]*x^i*y^j, i=0..dB[I]-j), j=0..min(dB
[I],10*d-1)): od:

for l from 0 to m - 2*3 do
CC||I := add(add(C||I[i,j]*x^i*y^j, i=0..dC[I]-j), j=0..min(dC
[I],10*d-1)): od:

for l from 0 to m - 3*3 do
DD||I := add(add(D||I[i,j]*x^i*y^j, i=0..dD[I]-j), j=0..min(dD
[I],10*d-1)): od:

for l from 0 to m - 4*3 do
EE||I := add(add(E||I[i,j]*x^i*y^j, i=0..dE[I]-j), j=0..min(dE
[I],10*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(10*d)

Systemey10d := {}:

for k from 0 to (m-10)/3 do

Coeffs_y10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10
-3*k,k], y, 10*d)), [x,y])}:

Systemey10d := Systemey10d union Coeffs_y10d[10,m-10-3*k,k]:

od:

Systeme_y10d := Systemey10d:

## VARIABLES FOR DIVISIBILITY BY y^(10*d)

Variables_y10d := indets(Systeme_y10d):

## RESOLUTION y^(10*d)

Solutions_y10d := solve(Systeme_y10d, Variables_y10d):

assign(Solutions_y10d):

##

for l from 0 to m - 0*3 do nops(expand(AA||I)) od;
for l from 0 to m - 1*3 do nops(expand(BB||I)) od;

```

```

for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EEl||l)) od;

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(11)

>

```

## EXPAND ALL UNKNOWNNS UP TO y^(11*d)

```

```

for l from 0 to m - 0*3 do

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```

AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],11*d-1)): od:

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for l from 0 to m - 1*3 do

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BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],11*d-1)): od:

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for l from 0 to m - 2*3 do

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CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],11*d-1)): od:

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```

for l from 0 to m - 3*3 do

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DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],11*d-1)): od:

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for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],11*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(11*d)

Systemey11d := {}:

for k from 0 to (m-11)/3 do

Coeffs_y11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11
-3*k,k], y, 11*d)), [x,y])}:

Systemey11d := Systemey11d union Coeffs_y11d[11,m-11-3*k,k]:

od:

Systeme_y11d := Systemey11d:

## VARIABLES FOR DIVISIBILITY BY y^(11*d)

Variables_y11d := indets(Systeme_y11d):

## RESOLUTION y^(11*d)

Solutions_y11d := solve(Systeme_y11d, Variables_y11d):

assign(Solutions_y11d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

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(12)

```
> ## EXPAND ALL UNKNOWNNS UP TO y^(12*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],12*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],12*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],12*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],12*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],12*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(12*d)

Systemey12d := {}:

for k from 0 to (m-12)/3 do

Coeffs_y12d[12,m-12-3*k,k] := {coeffs(expand(mtaylor(Eq[12,m-12
-3*k,k], y, 12*d)), [x,y])}:

Systemey12d := Systemey12d union Coeffs_y12d[12,m-12-3*k,k]:
```

od:

Systeme_y12d := Systemey12d:

VARIABLES FOR DIVISIBILITY BY $y^{(12*d)}$

Variables_y12d := indets(Systeme_y12d):

RESOLUTION $y^{(12*d)}$

Solutions_y12d := solve(Systeme_y12d, Variables_y12d):

assign(Solutions_y12d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EF||l)) od;

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(13)

> ## EXPAND ALL UNKNOWNNS UP TO $y^{(13*d)}$

for l from 0 to m - 0*3 do

AA||l := add(add(A||l[i,j]*xⁱy^j, i=0..dA[l]-j), j=0..min(dA

```

[l],13*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],13*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],13*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],13*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],13*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(13*d)

Systemey13d := {}:

for k from 0 to (m-13)/3 do

Coeffs_y13d[13,m-13-3*k,k] := {coeffs(expand(mtaylor(Eq[13,m-13
-3*k,k], y, 13*d)), [x,y])}:

Systemey13d := Systemey13d union Coeffs_y13d[13,m-13-3*k,k]:

od:

Systeme_y13d := Systemey13d:

## VARIABLES FOR DIVISIBILITY BY y^(13*d)

Variables_y13d := indets(Systeme_y13d):

## RESOLUTION y^(13*d)

Solutions_y13d := solve(Systeme_y13d, Variables_y13d):

assign(Solutions_y13d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;

```



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for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

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> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(14*d)}$ 

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```

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],14*d-1)): od:

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```

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],14*d-1)): od:

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```

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],14*d-1)): od:

```

```

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],14*d-1)): od:

```

```

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],14*d-1)): od:

```

```

## SYSTEM FOR DIVISIBILITY BY  $y^{(14*d)}$ 

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```

Systemey14d := {}:

for k from 0 to (m-14)/3 do

Coeffs_y14d[14,m-14-3*k,k] := {coeffs(expand(mtaylor(Eq[14,m-14
-3*k,k], y, 14*d)), [x,y])}:

Systemey14d := Systemey14d union Coeffs_y14d[14,m-14-3*k,k]:

od:

Systeme_y14d := Systemey14d:

## VARIABLES FOR DIVISIBILITY BY y^(14*d)

Variables_y14d := indets(Systeme_y14d):

## RESOLUTION y^(14*d)

Solutions_y14d := solve(Systeme_y14d, Variables_y14d):

assign(Solutions_y14d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

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> ## HOLOMORPHICITY AT INFINITY

  for l from 0 to m - 0*3 do

    HA||l := add(add(A||l[i,j]*X^i*Y^j*T^(dA[l]-i-j), i=0..dA[l]-j),
j=0..dA[l]):

    od:

    #

    for l from 0 to m - 1*3 do

      HB||l := add(add(B||l[i,j]*X^i*Y^j*T^(dB[l]-i-j), i=0..dB[l]-j),
j=0..dB[l]):

      od:

      #

      for l from 0 to m - 2*3 do

        HC||l := add(add(C||l[i,j]*X^i*Y^j*T^(dC[l]-i-j), i=0..dC[l]-j),

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j=0..dC[l]):

od:

#

for l from 0 to m - 3*3 do

HD[l] := add(add(D[l][i,j]*X^i*Y^j*T^(dD[l]-i-j), i=0..dD[l]-j),
j=0..dD[l]):

od:

#

for l from 0 to m - 4*3 do

HE[l] := add(add(E[l][i,j]*X^i*Y^j*T^(dE[l]-i-j), i=0..dE[l]-j),
j=0..dE[l]):

od:

##

for j from 0 to m - 0*3 do

HK[j,0] := HA[j]:

od:

#

for j from 0 to m - 1*3 do

HK[j,1] := HB[j]:

od:

#

for j from 0 to m - 2*3 do

HK[j,2] := HC[j]:

od:

#

```

```
for j from 0 to m - 3*3 do
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```
HK[j,3] := HD||j:
```

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od:
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#
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```
for j from 0 to m - 4*3 do
```

```
HK[j,4] := HE||j:
```

```
od:
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```
> ##
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```
printlevel := 2:
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```
for q from 0 to m/3 do
```

```
for p from 0 to m-3*q do
```

```
HEq[p,q] := expand(mtaylor(add(add(add(
```

```
    (-1)^k * 2^(p-r) * (d+1)^(j+k-p-q) * binomial(j,r) * (k!)/((p-  
r)!*q!*(k+r-p-q)!)
```

```
    * X^(m-q-j-2*k) * T^(k-q) * RY^(2*k-2*q) * HK[j,k],
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    j=r..m-3*k),
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    r=max(0,p+q-k)..min(p,m-3*k)),
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    k=q..m/3)
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, T, m-p-3*q)):
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od: od:
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```
> HSysteme := {seq(seq(coeffs(HEq[p,q],[T,X,Y]),p=0..m-3*q),q=0..  
m/3)} minus {0}:
```

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HSolutions := solve(HSysteme):
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assign(HSolutions):
```

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##
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```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
```

```
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
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```
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
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for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
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```
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
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> Jet_xR := factor(
  + (1/Ry^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) *
  (R1/RR)^j * Delta_xR^0, j=0..m-0*3)
  + (1/Ry^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) *
  (R1/RR)^j * Delta_xR^1, j=0..m-1*3)
  + (1/Ry^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) *
  (R1/RR)^j * Delta_xR^2, j=0..m-2*3)
  + (1/Ry^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) *
  (R1/RR)^j * Delta_xR^3, j=0..m-3*3)
  + (1/Ry^(m-4*2)) * x^(-twist) * add(EE||j * x1^(m-4*3-j) *
  (R1/RR)^j * Delta_xR^4, j=0..m-4*3) ):

```

```

> Vars := indets(Jet_xR) minus {x, x1, x2, y, y1, y2, R1, R2};
  NbVars := nops(Vars);

  Vars:= { }
  NbVars:= 0

```

(17)

```

> Ens_Vars := {}:

```

```

for i from 1 to nops(Vars) do
  Ens_Vars := Ens_Vars union {op(i,Vars)=0}:
od:
EnsVars := Ens_Vars:

```

```

> ## SOLUTIONS WITH FORMAL R1 AND R2

for i from 1 to nops(Vars) do
  JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jet_xR))):
od:

```

```

> ## EXPLICIT SOLUTIONS

R1 := x1*Rx + y1*Ry:
R2 := x2*Rx + y2*Ry + x1^2*Rxx + 2*x1*y1*Rxy + y1^2*Ryy:
Delta_xR := factor(x1*(R2/RR-R1^2/RR^2) - x2*(R1/RR)):

##

for i from 1 to nops(Vars) do
  JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jet_xR))):
od:

```

```

> ## VERIFICATION OF HOLOMORPHICITY AT INFINITY

```



```

Liste_x := {x = 1/x,
            y = y/x,
            x1 = - x1/x^2,
            x2 = 2*x1^2/x^3 - x2/x^2,
            y1 = y1/x - y*x1/x^2,
            y2 = y2/x - 2*y1*x1/x^2 + 2*y*x1^2/x^3 - y*x2/x^2}:

#

Liste_y := {y = 1/y,
            x = x/y,
            y1 = - y1/y^2,
            y2 = 2*y1^2/y^3 - y2/y^2,
            x1 = x1/y - x*y1/y^2,
            x2 = x2/y - 2*x1*y1/y^2 + 2*x*y1^2/y^3 - x*y2/y^2}:

##

for i from 1 to nops(Vars) do

JetSol_x[i] := factor(subs(Liste_x, JetSol[i]));
JetSol_y[i] := factor(subs(Liste_y, JetSol[i]));

od:

```

```

> Vars := indets(Jet_xR) minus {x, x1, x2, y, y1, y2, R1, R2};

NbVars := nops(Vars);

```

```

Vars:= { }
NbVars:= 0

```

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